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The Essentials

Why Study High-Dimensional Spaces?

Data Science studies how to represent the real world in the form of numerical data within mathematical spaces. It focuses on the **geometric** and **statistical** properties of these spaces.

Universal Pipeline:

- Real world → Sensors → Raw data → Numerical representation
- Everything becomes a **matrix**: N samples, D features

Two Complementary Perspectives:

- **Statistical:** Sample from an underlying distribution

- **Algebraic:** Matrix $X \in \mathbb{R}^{D \times N}$

The Representation Space

A D-dimensional space has a complete structure:

- Inner product $\rightarrow$ Norm $\rightarrow$ Distance $\rightarrow$ Metric space

Central Question: What happens when D becomes large?

- Negative effects: **Curse** of dimensionality
- Positive effects: **Blessing** of dimensionality

The Curse of Dimensionality

1. Data Sparsity

To accurately estimate a distribution in D dimensions, an **exponential** number of samples is required:

$$N \approx n \cdot b^D$$

where n = samples per bin, b = number of bins per dimension.

Example:

- $D = 6$, $b = 10$, $n = 10 \rightarrow N \approx 10^7$ samples needed
- With fixed N: most bins remain **empty**

Details: Estimation Quality

If N is fixed, the estimation quality becomes:

$$n = \frac{N}{b^D} \implies \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b^D}$$

The probability that a bin is occupied decreases exponentially with D.

Quick Review: Data Sparsity

Requires $N \approx n \cdot b^D$ samples (**exponential** growth).
If N is fixed: $n = N/b^D \rightarrow$ most bins **empty**.

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2. Empty Sphere and Spiky Cube

In a high-dimensional cube, the volume of the inscribed sphere becomes **negligible**:

$$\frac{\mathrm{vol}(B_D(r))}{\mathrm{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1}\Gamma(D/2)} \xrightarrow{D\to\infty} 0$$

Consequences:

- Volume concentrates in the **corners** of the cube
- Center-corner distance = $\frac{\sqrt{D}}{2}$ (grows with $\sqrt{D}$)
- Uniform sampling → **empty** sphere

Details: Volume of the Hypersphere

General Formula:

$$\mathrm{vol}(B_D(r)) = \frac{2r^D\pi^{D/2}}{D\Gamma(D/2)}$$

Recurrence Relation (for $D > 1$):

$$\mathrm{vol}(B_D(r)) = \frac{2\pi r^2}{D}\mathrm{vol}(B_{D-2}(r))$$

Corners of the Unit Cube:

- Number of corners: 2^D
- Distance to origin: $\|x_i\|_2 = \frac{\sqrt{D}}{2}$

D	1	2	4	5	100
$\frac{\sqrt{D}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{5}}{2}$	5

Quick Review: Empty Sphere & Spiky Cube

Sphere/cube ratio $\to 0$. Volume concentrated at **corners** (distance $\sqrt{D}/2$).
Uniform sampling $\to$ **empty** sphere.

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3. Volume Distribution

For **any shape** (cube, sphere, or other), volume concentrates near the surface:

$$\lim_{D\to\infty} \frac{\mathrm{vol}(V_D(r)) - \mathrm{vol}(V_D((1-\varepsilon)r))}{\mathrm{vol}(V_D(r))} = 1$$

Principle:

- Reducing size by factor $(1-\varepsilon) \to$ volume reduced by $(1-\varepsilon)^D \to 0$
- Almost all volume is in a thin **shell** at the surface

Details: Calculation for Hypercube and Hypersphere

For the Hypercube:

$$\frac{\mathrm{vol}(C_D((1-\varepsilon)r))}{\mathrm{vol}(C_D(r))} = (1-\varepsilon)^D \xrightarrow{D\to\infty} 0$$

For the Hypersphere:

$$\frac{\mathrm{vol}(B_D((1-\varepsilon)r))}{\mathrm{vol}(B_D(r))} = (1-\varepsilon)^D \xrightarrow{D\to\infty} 0$$

Technical Note: $(1-\varepsilon)^D \le e^{-D\varepsilon} \to 0$ as $D \to \infty$
Decomposition Method: Divide the volume into arbitrarily small hypercubic voxels, then apply the same reasoning.

Quick Review: Volume Distribution

For **any shape:** ratio $(1-\varepsilon)^D \to 0$.
Volume **concentrated at the surface**.

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4. The Gaussian Egg

For a distribution $X \sim \mathcal{N}(0, \mathrm{Id}_D)$, the radius $R = \|X\|_2$ follows:

$$R^2 = \sum_{i=1}^{D} X_i^2 \sim \chi^2(D) \implies \mathbb{E}[R^2] = D \implies \mathbb{E}[R] \approx \sqrt{D}$$

The density is **concentrated** around the radius $\sqrt{D}$.

Gaussian Annulus Theorem:

$$P\left[\left|\|X\|_2 - \sqrt{D}\right| > \beta\right] \leq 3e^{-c\beta^2}$$

Almost all density is in a **thin ring** around $\sqrt{D}$.

Details: Density Along the Radius

Context: Coordinates $X_i \sim \mathcal{N}(0, 1)$ i.i.d.
Integration along r = take the norm. R is the random variable of the radius.
Alternative Formulation:

$$P\left[\left|\frac{\|X\|}{\sqrt{D}} - 1\right| \leq \varepsilon\right] \approx 1$$

For a Gaussian $\mathcal{N}(0, \sigma^2 \mathrm{Id}_D)$, the density is concentrated at radius $\sigma\sqrt{D}$.

Quick Review: Gaussian Egg

For $X \sim \mathcal{N}(0, \mathrm{Id}_D)$: $R^2 \sim \chi^2(D)$ so $\mathbb{E}[R^2] = D$.
Density concentrated at radius $\sqrt{D}$ (**thin ring**).

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5. Distance Concentration

All neighbors become **equidistant** as D increases.

Theorem: If the variance of the normalized distance tends to 0, then:

$$\lim_{D \to \infty} P\left[\frac{D_{\max} - D_{\min}}{D_{\min}} \leq \varepsilon\right] = 1$$

where $D_{\min}$ and $D_{\max}$ are the distances to the nearest and farthest neighbors.

Consequences:

- All neighbors appear at distance D_{min}
- Discriminative power decreases exponentially
- KNN and ε-NN become **non-significant**

Details: Concentration Condition

Complete Theorem: If $\lim_{D\to\infty} \text{Var}\left[\frac{d(x,y)^p}{\mathbb{E}[d(x,y)^p]}\right] = 0$ for $0 < p < \infty$, then concentration occurs.
This condition means that the relative fluctuations of the distance become negligible.

Quick Review: Distance Concentration

If normalized distance variance $\to 0$, then $\frac{D_{\text{max}} - D_{\text{min}}}{D_{\text{min}}} \to 0$.
All neighbors **equidistant**. KNN loses meaning.

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6. Quasi-Orthogonality

In high dimensions, random vectors exhibit surprising geometric properties:

Two random vectors are quasi-orthogonal:

$$\mathbb{E}[\cos(\theta)] = 0 \implies \theta \approx \frac{\pi}{2}$$

Random triangle is quasi-equilateral:

$$\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \approx \frac{\pi}{3}$$

Consequence: On a sphere, volume concentrates at the **equator** (regardless of direction).

Details: Angle Calculation Between Vectors

For two random vectors:
Define 4 i.i.d. random variables W, X, Y, Z on $\Omega \subset \mathbb{R}^D$:

$$\mathbb{E}\left[\cos(\angle(X-Y, Z-W))\right] = \mathbb{E}\left[\frac{\langle X-Y, Z-W\rangle}{\|X-Y\| \cdot \|Z-W\|}\right]$$

Since W, X, Y, Z are i.i.d., $X - Y$ and $Z - W$ are also i.i.d. and centered: $\mathbb{E}(X - Y) = \mathbb{E}(Z - W) = 0$.
The numerator is $\text{cov}(X - Y, Z - W) = 0$.

For a triangle:
With 3 i.i.d. random variables X, Y, Z, compute:

$$\frac{\langle X-Y, Z-Y\rangle}{\langle X-Y, X-Y\rangle} = \frac{\langle X, Z\rangle - \langle X, Y\rangle - \langle Y, Z\rangle + \langle Y, Y\rangle}{\langle X, X\rangle + \langle Y, Y\rangle - 2\langle X, Y\rangle}$$

In expectation: $= \frac{1}{2}$
Note: To sample on the unit hypersphere, sample a centrally symmetric distribution (e.g., Gaussian) and normalize.

Quick Review: Quasi-Orthogonality

2 random vectors: $\theta \approx \pi/2$ (quasi-orthogonal).
Random triangle: $\theta \approx \pi/3$ (equilateral).
Sphere: volume at the equator.

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7. Hubness

Some samples appear in the neighborhood of **many** others.

Definition: $n_k(x_j)$ = number of times x_j is a k-NN of another point.

Observation: The distribution of n_k is **asymmetric**:

- A few points (**hubs**) have very high values
- Most points have low values
- These hubs are "close" to all other samples

Details: Formal Definition of Hubness

Given X, define:

$$r_{ik}(x_j) = \begin{cases} 1 & \text{if } x_j \in V_X^k(x_i) \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$n_k(x_j) = \sum_i r_{ik}(x_j)$$

Empirical Visualization: For 20-NN with $D = 100$ and 1000 uniform samples on 50 bins, a strong asymmetry in the distribution of n_k is observed.

Quick Review: Hubness

$n_k(x_j)$ = number of times x_j is a k-NN of another point.
Asymmetric distribution: few **hubs** in many neighborhoods.

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8. Concentration Inequalities

Concentration inequalities mathematically explain these phenomena.

Fundamental Inequalities:

- **Markov:** $P(X > \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon}$
- **Chebyshev:** $P(|X - \mu| > \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$

Hoeffding's Inequality: For $\{X_i\}_{i \in [\![D]\!]}$ independent with $P(a \leq X_i \leq b) = 1$ and $\mathbb{E}[X_i] = \mu$:

$$P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$$

where $\bar{X} = \frac{1}{D}\sum_{i=1}^{D} X_i$

Explanation:

- Aggregation of independent variables $\to$ convergence to μ
- Minkowski distances have structure $\sum_{i=1}^{D}(\cdot)$
- For $d_p(x, y)$ to be small (large), **all** D coordinates must be similar (dissimilar)
- As D increases, this becomes unlikely

Details: Application to Distances

Minkowski distances:

$$d_p(x, y) = \left(\sum_{i=1}^{D} |x[i] - y[i]|^p\right)^{1/p}$$

Hoeffding's inequality formally explains why distance values **concentrate**.
Intuition: For large D, extreme values of $d_p(x, y)$ become unlikely because they require coordination across **all** dimensions.
Note: Example of PAC (Probably Approximately Correct) analysis.

Quick Review: Concentration Inequalities

Hoeffding: $P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$.
Structure $\sum_{i=1}^{D}(\cdot)$ in distances $\to$ **concentration**.

The Blessing of Dimensionality

1. Laws of Large Numbers

Weak Law of Large Numbers (WLLN):

For $\{X_i\}_{i \in [\![D]\!]}$ i.i.d. with $\mathbb{E}[X_i] = \mu$:

$$\lim_{D \to \infty} P(|\bar{X} - \mu| < \varepsilon) = 1$$

where $\bar{X} = \frac{1}{D} \sum_{i=1}^{D} X_i$

Interpretation:

- The empirical mean $\bar{X}$ converges to the true expectation μ
- Fundamental basis of estimation theory
- Consistent with the frequentist approach to probabilities

Central Limit Theorem (CLT):

$$Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$$

if $\text{Var}(X_i) = \sigma^2$

Details: Proof via Chebyshev

The WLLN can be proven from Chebyshev's inequality:

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \frac{\text{Var}(\bar{X})}{\varepsilon^2} = \frac{\sigma^2}{D\varepsilon^2} \xrightarrow{D \to \infty} 0$$

The CLT gives the form (Gaussian) and the rate ($1/\sqrt{D}$) of convergence.

Quick Review: Laws of Large Numbers

WLLN: $\bar{X} \xrightarrow{P} \mu$ as $D \to \infty$.

Basis for density estimation.
CLT: convergence rate $1/\sqrt{D}$.

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2. Random Projections

Principle: Exponentially many quasi-orthogonal directions in high dimensions.

Johnson-Lindenstrauss Lemma:

Given $X \subset \Omega$, there exists a linear projection $p : \mathbb{R}^D \to \mathbb{R}^d$ such that:

$$(1-\varepsilon)\|x_i - x_j\| \leq \|p(x_i) - p(x_j)\| \leq (1+\varepsilon)\|x_i - x_j\|$$

with

$$d = O\left(\frac{\log N}{\varepsilon^2}\right)$$

Important Note: d does **NOT** depend on D!

Applications:

- Locally Sensitive Hashing (LSH) for approximate nearest neighbor (ANN) search
- Random bases for signal decomposition and coding

Caveat: Convergence rate slower than most concentrations

Details: Why It Works

Recall: Two random vectors u and v are likely orthogonal: $\langle u, v\rangle = \varepsilon$ (small).
Key Property: There are exponentially many quasi-orthogonal directions in high-dimensional spaces.
This allows creating random bases (quasi-orthonormal) that preserve essential metric properties while drastically reducing dimension.

Quick Review: Random Projections

Johnson-Lindenstrauss: projection $\mathbb{R}^D \to \mathbb{R}^d$ preserving distances with $d = O(\log N/\varepsilon^2)$ (independent of D).
Application: LSH for ANN.

Impact in Data Science

1. Sampling

Consequences of Sparsity:

- Requires an **exponential** number of samples to cover the space
- Concentration inequalities are **precision indicators** for estimation
- Rejection rate in rejection sampling $\rightarrow$ close to 100%

Details: Rejection Sampling

In practical processes like rejection sampling, the rejection rate becomes exponentially close to 100% due to the empty sphere phenomenon.
If sampling in a particular region (like a sphere inscribed in a cube), the acceptance probability decreases exponentially with D.

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2. Indexing

Exploiting Structure:

- Indexing exploits data structure to anticipate which parts of the space to visit
- If all neighbors are at equal distance, the structure **disappears**

Spatial Partitioning Techniques (e.g., kd-trees):

- Attempt to uniquely identify each data point in a cell
- With distance concentration: cell size $\rightarrow$ full radius
- Search becomes **exhaustive** $\rightarrow$ complexity $O(N)$
- Hubness makes some data points answers to almost all queries

Details: Search Trees

kd-trees attempt to partition the space so that each data point is uniquely identified within a cell intersection (paths in the tree).
With distance concentration:

- The cell must be as large as the sphere's radius to cover the entire space
- Pruning conditions become ineffective
- The tree degenerates into a linear list

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3. Learning

Machine Learning = discovering function approximators.

Problem: Find $\phi_\theta \in \mathcal{F}$ approximating $\phi : X \to Y$ by minimizing an error.

Catastrophic Theoretical Bound:

If $\mathcal{F}$ is a class of c-uniformly Lipschitz functions on Ω:

$$\sup_{\phi_\theta \in \mathcal{F}} \|\phi_\theta - \phi\|_\infty \geq \frac{c\sqrt{D}N^{-1/D}}{2}\sqrt{\frac{2}{\pi e}}\left(1 + O\left(\frac{\log D}{D}\right)\right)$$

For an error $c\varepsilon$ with $\varepsilon > 0$, the required number of examples is:

$$N \geq \frac{\varepsilon^{-D} D^{D/2}}{(2\pi e)^{D/2}}$$

Concrete Examples:

- $D = 3$, $\varepsilon = 0.1 \to N \geq 74$
- $D = 10$, $\varepsilon = 0.1 \to N \geq 687 \times 10^6$
- $D = 15$, $\varepsilon = 0.1 \to N \geq 377 \times 10^{12}$
- $D = 10$, $\varepsilon = 0.01 \to N \geq 6.9 \times 10^{18}$

Realistic Case: $D \sim 100$, $\varepsilon \sim 10^{-4} \to N \sim 10^{400}$ **IMPOSSIBLE**

> **Quick Review: Importance in Data Science**
>
> **Sampling:** Exponential need, rejection rate $\to$ 100%.
> **Indexing:** Structure disappears, kd-trees $\to O(N)$, hubness.
> **Learning:** $N \geq \varepsilon^{-D} D^{D/2}/(2\pi e)^{D/2} \to$ impossible for large D.

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4. Why Does Data Science Still Work?

Assumptions in the Above Analyses:

- **Uniform** (or isotropic) data distribution
- **Independent** features

These assumptions are FALSE in practice!

Reality:

- Real data **do NOT** populate the space uniformly
- Features show **correlations**
- Data occupy **subspaces** of lower intrinsic dimensionality

Practical Solutions:

- Decorrelate features (linear latent space)
- Discover subspaces (clustering, nonlinear dimensionality reduction)
- PCA, autoencoders, neural networks
- Exploit underlying geometric structure

Key Insight: If features are informative, they are not independent, and data concentrate on manifolds of much lower dimension than D.

> **Quick Review: Why It Works**
>
> Analyses assume **uniform** distribution and **independent** features.
> **Reality:** Data lie on **subspaces** of low intrinsic dimensionality.
> **Solutions:** Decorrelation, clustering, dimensionality reduction.

Global Summary

High-dimensional geometry ($D \gtrsim 10$) is **different** from familiar 3D geometry.

Positive Aspects (Blessing)

- **Concentration inequalities:** Basis of statistical estimation
- **Laws of Large Numbers:** Enable density estimation
- **Random projections:** Exploit quasi-orthogonality

Negative Aspects (Curse)

- Data highly **sparse** (exponential sample requirement)
- **Concentration of distances and angles:**
  - All neighbors equidistant
  - Everything appears orthogonal
- Density concentrated on **narrow radii** $\rightarrow$ biased sampling
- **Hubs:** Part of all classes/responses

Unique Solution

Since this is by construction, the only solution is to operate in **low-dimensional spaces:**

- Dimensionality reduction
- Subspace discovery
- Feature decorrelation

Key Message: Data Science works because real data **are not uniform** and live on **low-dimensional submanifolds**.

Quick Review: Summary

High dimension ($D \gtrsim 10$) 3D geometry.
: Concentration inequalities, WLLN, random projections.
: Data sparsity, distance/angle concentration, hubs.
Solution: Operate in **low dimension** (PCA, autoencoders, structure exploitation).